25ECPCC303IAE-II 03 OCT 2026

Seat No. : 15

Student Name : Harsh

St. John College of Engineering and Management, Palghar,
Maharashtra
Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC303 - Digital Electroncs and Comp Architecture

11-12

Time : 1 Hour **Maximum Marks : 20**

St. John College of Engineering & Management SJCEM PALGHAR

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Maximum marks(5)	Course Outcome	RBT Level
	a)What is the feature of CPLD	1	CO4	L1
	b)What is the adavantages of FPGA?	1	CO4	L1
	c)Define any four oprators in VHDL	1	CO5	L1
	d)Whar are the advantages of HDL?	1	CO5	L1
	e)What does the term "fetch" refer to in the instruction cycle?	1	CO6	L1
	f)Explain the concept of virtual memory.	1	CO6	L2
Q2	a)Write the data flow Mode VHDL code for EX-OR GATE	3 5	CO4	L3
	OR			
	b)Write the structural mode VHDL code for Half Adder	3 5	CO4	L3
Q3	a)Draw and explain instruction cycle.	4 5	CO5	L3
	OR			
	b)Draw and explain CPU architecture .	5	CO5	L3
Q4	a)Define primary memory and secondary memory with examples.	5	CO6	L2
	OR			

	b)Describe various cache memory mapping techniques.	5	CO6	L3

St. John College of Engineering & Management
SJCEM
PALGHAR